

Task Report

Fine-Tuning DistilBERT for Sentiment Analysis

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Natural Language Processing | IMDB Sentiment Classification

1. Introduction

Natural Language Processing (NLP) has witnessed remarkable progress with the advent of transformer-based models. Among these, BERT (Bidirectional Encoder Representations from Transformers) and its variants have set new benchmarks across a wide range of text classification tasks. This project focuses on fine-tuning DistilBERT — a lightweight and computationally efficient distillation of BERT — for the task of binary sentiment classification using the IMDB movie review dataset.

Sentiment analysis is the process of identifying and categorizing opinions expressed in a piece of text, determining whether the expressed sentiment is positive, negative, or neutral. In this work, we perform binary classification (positive vs. negative) to evaluate how well DistilBERT can capture emotional tone from natural language text.

1.1 What is DistilBERT?

DistilBERT is a smaller, faster, and lighter version of BERT, developed by Hugging Face. It retains approximately 97% of BERT's language understanding capability while being 40% smaller and 60% faster. This makes it ideal for production environments and academic experiments where computational resources are limited.

Key properties of DistilBERT:

- 6 transformer layers (vs. 12 in BERT-base)
- 66 million parameters (vs. 110 million in BERT-base)
- Pre-trained using knowledge distillation on the same corpus as BERT
- Retains BERT's tokenizer and attention mechanism

1.2 Objective

The primary objective of this project is to fine-tune a pre-trained DistilBERT model on a downstream sentiment classification task and evaluate its performance using standard NLP metrics including accuracy, precision, recall, and F1-score.

2. Dataset

The IMDB Movie Review Dataset is a widely used benchmark for binary sentiment classification in NLP research. It was introduced by Maas et al. (2011) and consists of 50,000 movie reviews scraped from the Internet Movie Database (IMDb).

2.1 Dataset Overview

Total Reviews	50,000
Training Set	25,000 reviews
Test Set	25,000 reviews
Classes	Binary — Positive / Negative
Class Balance	Perfectly balanced (50% each class)
Average Review Length	~230 words per review
Source	Internet Movie Database (IMDb)

2.2 Data Characteristics

Each review is labeled as either positive (score ≥ 7 out of 10) or negative (score ≤ 4 out of 10). Reviews with neutral scores (5–6) were excluded to ensure label clarity. The dataset is perfectly balanced, with 25,000 positive and 25,000 negative reviews, which eliminates class imbalance bias during training and evaluation.

For this project, a subset of 2,000 reviews (1,000 per class) was used for evaluation after fine-tuning the model on the full training split. This evaluation set provides a reliable and unbiased measure of model generalization.

3. Methodology

The fine-tuning pipeline follows a standard transfer learning approach for NLP. A pre-trained DistilBERT model is adapted to the sentiment classification task by adding a classification head and training it on labeled IMDB data.

3.1 Model Architecture

The base model used is distilbert-base-uncased from the Hugging Face Transformers library. A linear classification head is appended on top of the [CLS] token representation to produce binary logits. The architecture can be summarized as:

- Input: Tokenized text sequences (max length: 512 tokens)
- Encoder: 6-layer DistilBERT transformer with 768 hidden dimensions
- Pooling: [CLS] token output (first token representation)
- Classifier Head: Linear(768 → 2) followed by softmax activation
- Output: Probability scores for Positive and Negative classes

3.2 Tokenization

Text is tokenized using the DistilBertTokenizer from Hugging Face. The tokenizer applies WordPiece subword tokenization, converting raw text into token IDs with corresponding attention masks. Special tokens [CLS] and [SEP] are automatically added to mark the beginning and end of each sequence.

3.3 Training Configuration

Base Model	distilbert-base-uncased (Hugging Face)
Task	Sequence Classification (Binary)
Optimizer	AdamW
Learning Rate	2×10^{-5}
Number of Epochs	3
Batch Size	16 (train) / 16 (eval)
Max Sequence Length	512 tokens
Loss Function	Cross-Entropy Loss
Weight Decay	0.01 (regularization)
Warmup Steps	500
Framework	PyTorch + Hugging Face Transformers

3.4 Training Strategy

The model was trained using the Hugging Face Trainer API, which abstracts the training loop and provides built-in support for evaluation callbacks, gradient accumulation, and mixed precision training. The AdamW optimizer was chosen for its improved weight decay behavior compared to standard Adam, which helps reduce overfitting in transformer models.

A linear learning rate warmup schedule was applied during the first 500 steps, gradually increasing the learning rate from 0 to 2×10^{-5} before maintaining it constant for the remainder of training. This approach stabilizes early training and prevents large gradient updates from destabilizing the pre-trained weights.

4. Training Results

The model was evaluated at the end of each epoch on a held-out validation set. Training and validation metrics were recorded to monitor model convergence and detect potential overfitting.

4.1 Epoch-wise Performance

Epoch	Training Loss	Validation Loss	Accuracy	F1 Score
1	0.3139	0.2289	91.15%	0.9122
2	0.1713	0.2744	90.95%	0.9112
3	0.1023	0.3308	90.95%	0.9109

4.2 Analysis of Training Dynamics

Several important patterns emerge from the training curve:

- **Training Loss:** Consistently decreases across all three epochs (0.3139 → 0.1713 → 0.1023), indicating that the model is successfully learning patterns from the training data.
- **Validation Loss:** Increases after Epoch 1 (0.2289 → 0.2744 → 0.3308), suggesting that the model begins to slightly overfit to the training set from Epoch 2 onwards.
- **Accuracy:** Peaks at Epoch 1 (91.15%) and stabilizes around 90.95% for subsequent epochs, confirming that Epoch 1 yields the most generalizable checkpoint.
- **F1 Score:** Follows a similar trend to accuracy — highest at Epoch 1 (0.9122) with marginal decline in later epochs.

The divergence between training and validation loss is a hallmark of mild overfitting. However, since validation accuracy and F1 remain stable (not declining significantly), the overfitting is considered benign in this context. The optimal model checkpoint corresponds to Epoch 1, and in practice, early stopping would be recommended here.

5. Evaluation Results

After training, the best model checkpoint (Epoch 1) was evaluated on a held-out test set of 2,000 reviews (1,000 positive and 1,000 negative). The following per-class and aggregate metrics were computed.

5.1 Classification Report

Class	Precision	Recall	F1-Score	Support
Negative	0.92	0.90	0.91	1,000
Positive	0.91	0.92	0.91	1,000
Accuracy	—	—	0.91	2,000
Macro Average	0.91	0.91	0.91	2,000
Weighted Average	0.91	0.91	0.91	2,000

5.2 Metric Definitions

- Precision: Of all reviews predicted as a given class, the fraction that were actually that class. High precision means fewer false positives.
- Recall: Of all reviews that actually belong to a given class, the fraction that were correctly predicted. High recall means fewer false negatives.
- F1-Score: The harmonic mean of precision and recall. Provides a single balanced metric, especially useful when classes are equally important.
- Support: The number of actual instances of each class in the test set.

5.3 Key Observations

The evaluation results demonstrate strong and consistent performance:

- Both classes achieve equal F1-scores of 0.91, indicating the model treats positive and negative sentiment with equal proficiency.
- Negative class precision (0.92) is slightly higher than positive class precision (0.91), suggesting the model is marginally more conservative when predicting negative sentiment.
- Positive class recall (0.92) is slightly higher than negative class recall (0.90), meaning the model is slightly better at detecting positive reviews from the full set.
- Perfect class balance in the test set (1,000 samples each) ensures that macro and weighted averages are identical, validating the fairness of the evaluation.

6. Discussion

The fine-tuned DistilBERT model achieves 91% accuracy and a macro F1-score of 0.91 on binary sentiment classification of IMDB movie reviews. This performance is competitive with much larger models, validating the effectiveness of transfer learning even with a distilled, compressed architecture.

6.1 Strengths

- **High Performance with Low Resources:** DistilBERT achieves near-BERT-level accuracy with 40% fewer parameters, making it practical for environments with limited compute.
- **Balanced Classification:** The model generalizes equally well across both sentiment classes, demonstrating no systematic class bias.
- **Rapid Convergence:** Peak validation performance is reached in just 1 epoch, indicating that DistilBERT's pre-trained representations are well-suited to this task and require minimal adaptation.
- **Stable F1-Score:** The F1-score remains above 0.91 throughout training, suggesting robustness to the mild overfitting observed in later epochs.

6.2 Limitations and Observations

- **Overfitting from Epoch 2:** The increasing validation loss from Epoch 2 onwards suggests the model begins to memorize training-specific patterns. Early stopping or stronger regularization could mitigate this.
- **Short Training Duration:** Only 3 epochs were run, which may not fully explore the model's potential. A learning rate scheduler with cosine annealing could improve final performance.
- **Domain Specificity:** The model is trained and evaluated on movie reviews. Performance may degrade on out-of-domain text (e.g., product reviews, social media posts) without domain adaptation.
- **Binary Classification Only:** The model does not capture fine-grained sentiment intensity (e.g., slightly positive vs. very positive). Multi-class or regression-based approaches could address this.

6.3 Recommendations for Improvement

- Implement early stopping based on validation F1-score to select the optimal checkpoint automatically.
- Experiment with data augmentation techniques (back-translation, synonym replacement) to improve generalization.
- Try learning rate scheduling (cosine annealing or polynomial decay) to achieve smoother convergence.
- Compare against full BERT-base and RoBERTa to quantify the accuracy trade-off of distillation.
- Evaluate on additional sentiment datasets (SST-2, Yelp) to test cross-domain robustness.

7. Conclusion

This project demonstrates the effectiveness of fine-tuning DistilBERT for binary sentiment analysis using the IMDB dataset. The model achieves a strong accuracy of 91% and a balanced F1-score of 0.91 across both positive and negative sentiment classes after just one epoch of training — a testament to the power of transfer learning in NLP.

The results confirm that DistilBERT, despite being a compressed variant of BERT, retains sufficient linguistic understanding to perform competitively on real-world sentiment classification tasks. The training dynamics reveal mild overfitting from Epoch 2 onwards, suggesting that early stopping would be a beneficial addition to the pipeline.

Overall, this work validates DistilBERT as a practical and efficient solution for sentiment analysis, particularly in resource-constrained environments. Future work should focus on cross-domain evaluation, multi-class sentiment detection, and comparison with larger transformer architectures.